Orbis Robotics

OR-6

Load, unload and deburr robot cell

Cell Maintenance Guide

Revision 3

For qualified maintenance personnel

1 Safety

1.1 General safety

Only qualified maintenance personnel who are trained on the OR-6 may enter the cell. Never bypass the light curtain or a door interlock, and stay clear of the arm: it can move without warning while the cell has power. If anything moves unexpectedly, press the emergency stop and leave the cell by the safety door.

1.2 Lockout and tagout

Move the robot to the service position, switch off the cell isolator and fit your personal lock and tag. The drive capacitors keep a dangerous charge after power off, so wait 5 minutes for the drive capacitors to discharge before you touch a drive. Release the compressed air from the gripper valve block and verify zero energy before you start.

2 Alarm table

The control shows these codes. Likely causes are listed in order of likelihood.

Code	Message	Severity	Likely causes	Action
RB-551	Gripper part-present signal lost	warning	worn gripper fingers (most likely)	See 4.1 Replace gripper fingers (plant ref PROC-RW-01)
RB-560	Axis 2 overcurrent	critical	axis 2 overload from binding or a failing drive (most likely)	See 4.2 Inspect axis 2 and reset servo drive (plant ref PROC-RW-02)
RB-563	Axis 2 following error	warning	axis 2 overload from binding or a failing drive (most likely)	See 4.2 Inspect axis 2 and reset servo drive (plant ref PROC-RW-02)
RB-570	Light curtain muting fault	critical	misaligned light curtain (most likely)	See 4.3 Realign light curtain (plant ref PROC-RW-03)

3 Troubleshooting table

Find the symptom, then follow the remedy.

Symptom	Probable cause	Remedy
robot dropping parts at the deburr station	worn gripper fingers	See 4.1 Replace gripper fingers (plant ref PROC-RW-01)
axis 2 trips on overcurrent during fast moves	axis 2 overload from binding or a failing drive	See 4.2 Inspect axis 2 and reset servo drive (plant ref PROC-RW-02)
cell stops randomly, light curtain fault	misaligned light curtain	See 4.3 Realign light curtain (plant ref PROC-RW-03)

4 Procedures

4.1 Replace gripper fingers

Plant ref: PROC-RW-01 | Lockout required: yes | Time: 40 min

WARNING: Apply lockout and tagout before starting. Verify zero energy.

1. Move the robot to the service position and apply lockout and tagout on the cell isolator.
2. Release air from the gripper valve block.
3. Unbolt the worn fingers and fit two GRP-FNG-12 fingers.
4. Remove the lockout and teach the part-present sensor on a master part.

Order no.	Description	Qty
GRP-FNG-12	Gripper finger, hardened	2

4.2 Inspect axis 2 and reset servo drive

Plant ref: PROC-RW-02 | Lockout required: yes | Time: 90 min

WARNING: Apply lockout and tagout before starting. Verify zero energy.

1. Apply lockout and tagout on the cell isolator and wait 5 minutes for the drive capacitors to discharge.
2. Check the axis 2 gearbox for binding by moving the arm by hand.
3. Replace the drive fuse with FUSE-SRV-10, and replace the idler bearing with BRG-6205-2RS if it feels rough.
4. Remove the lockout, reset the drive and run axis 2 at 10 percent speed.

Order no.	Description	Qty
FUSE-SRV-10	Servo drive fuse 10 A	1
BRG-6205-2RS	Deep groove ball bearing 6205-2RS	1

4.3 Realign light curtain

Plant ref: PROC-RW-03 | Lockout required: yes | Time: 30 min

WARNING: Apply lockout and tagout before starting. Verify zero energy.

1. Apply lockout and tagout on the cell isolator.
2. Check both light curtain brackets and replace a bent one with BRKT-LC-01.
3. Align the emitter and receiver until all beam LEDs are green.
4. Remove the lockout and run the muting test from the safety menu.

Order no.	Description	Qty
BRKT-LC-01	Light curtain mounting bracket	1

5 Spare parts list

Order spare parts by part number.

Part number	Description	Fits
BATT-ENC-36	Encoder backup battery 3.6 V	Axis 2 servo
BELT-HTD-8M	Timing belt HTD 8M	Axis 2 servo
BRG-6205-2RS	Deep groove ball bearing 6205-2RS	Axis 2 servo
BRKT-LC-01	Light curtain mounting bracket	Light curtain
CABLE-ENC-05	Encoder cable 5 m	Axis 2 servo
CUP-VAC-40	Vacuum suction cup 40 mm	Gripper
FUSE-SRV-10	Servo drive fuse 10 A	Axis 2 servo
GRP-FNG-12	Gripper finger, hardened	Gripper
RELAY-SAF-24	Safety relay 24 V	Light curtain